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import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, confusion_matrix,
classification_report
X = np.array([
    [2, 16, 64, 3000, 12],
    [3, 32, 128, 4000, 16],
    [4, 64, 128, 4500, 20],
    [6, 128, 256, 5000, 48],
    [8, 256, 512, 6000, 50],
    [2, 32, 64, 3500, 13],
    [3, 64, 128, 4200, 16],
    [4, 128, 256, 4800, 32],
    [6, 256, 512, 5500, 48],
    [8, 512, 1024, 6000, 64],
])
y = np.array([
    0, 1, 1, 2, 2,
    0, 1, 1, 2, 2,
    0, 0, 1, 2, 2,
    0, 1, 1, 2, 2
])
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
model = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("MOBILE PRICE PREDICTION")
print("-----")
print("\nActual Classes:")
print(y_test)
print("\nPredicted Classes:")
print(y_pred)
print("\nAccuracy:")
print(accuracy_score(y_test, y_pred))
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))
print("\nClassification Report:")
print(classification_report(y_test, y_pred))

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new_mobile = np.array([[6, 128, 256, 5000, 48]])
prediction = model.predict(new_mobile)
print("\nNew Mobile Price Range:")
print(prediction[0])
if prediction[0] == 0:
    print("Low Price Range")
elif prediction[0] == 1:
    print("Medium Price Range")
else:
    print("High Price Range")
MOBILE PRICE PREDICTION
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Actual Classes:
[0 2 2 1]

Predicted Classes:
[0 2 2 1]

Accuracy:
1.0

Confusion Matrix:
[[1 0 0]
[0 1 0]
[0 0 2]]

Classification Report:

Classification Report:					
	precision	recall	f1-score	support	
0	1.00	1.00	1.00	1	
1	1.00	1.00	1.00	1	
2	1.00	1.00	1.00	2	
accuracy			1.00	4	
macro avg	1.00	1.00	1.00	4	
weighted avg	1.00	1.00	1.00	4	

New Mobile Price Range:
2
High Price Range